Sanskriti

Kumarisanskriti889@gmail.com | LinkedIn | GitHub | Portfolio

Education

Scaler School of Technology **Aug 2024 – Present**
(In collaboration with BITS Pilani) ***CGPA: 9.05*** *Bengaluru, India*

Experience

Code-X-Novas **Sep 2025 – Nov 2025**
Software Developer Intern *Bengaluru, India*

- Developed responsive React.js frontend components for a company website serving 100+ users, translating UI/UX requirements into functional, user-friendly interfaces.
- Integrated frontend features with REST APIs and contributed to authentication configuration, supporting secure access and end-to-end application workflows.

Projects

AgentEval – AI Agent Evaluation Platform | Live
Technologies: Python, FastAPI, PostgreSQL, Redis Streams, Docker, AsyncIO

- Designed an AI agent evaluation platform to execute, monitor, and benchmark agent workflows in controlled environments with reproducible execution records.
- Implemented an asynchronous API and worker architecture using FastAPI, PostgreSQL for persistent state, and Redis Streams for task dispatch and delivery.
- Developed agent execution workflows with sandboxed tools and configurable step, time, token, and cost limits to control resource consumption.
- Implemented evaluation workflows with isolated execution environments, task validation, failure classification, execution traces, and structured scoring for benchmark comparisons.

RepoLens – Source Code Analysis Platform | Live
Technologies: Python, FastAPI, PostgreSQL, Tree-sitter, Docker, TypeScript

- Built a static code analysis platform that parses Python repositories and models functions, classes, imports, call relationships, inheritance, and test references.
- Implemented repository ingestion and immutable snapshots with file inventories, content hashes, language metadata, and unsafe-path detection.
- Developed static name resolution and a typed dependency graph with provenance and confidence labels to distinguish exact, heuristic, unresolved, and dynamic relationships.
- Implemented graph APIs for symbol search, caller and callee lookup, dependency traversal, shortest-path queries, import-cycle detection, and per-file analysis status.

KawaiiLM – AI-Powered Document Question Answering | Live
Technologies: Next.js, React, TypeScript, Express.js, Hugging Face, Qdrant, LangChain

- Built a retrieval-augmented generation (RAG) application for document question answering, with text extraction, chunking, and BAAI/bge-large-en-v1.5 embeddings stored in Qdrant for semantic search.
- Integrated Qwen2.5-7B-Instruct through Hugging Face Inference API and implemented SSE response streaming with page-level citations linked to a document viewer.

VisualSearch – Multimodal Semantic Image Retrieval | Live
Technologies: Python, PyTorch, CLIP, DINOv2, FAISS, Qdrant, FastAPI, React, ONNX Runtime

- Built a multimodal image retrieval system supporting natural-language, image-example, and hybrid queries with metadata filtering using CLIP and DINOv2 representations.
- Implemented vector search with FAISS indexes, score calibration, late fusion, and optional BLIP reranking to improve retrieval quality.
- Evaluated retrieval performance on reported benchmarks: CLIP ViT-L/14 achieved 64.7 R@1 on Flickr30k-1K, while BLIP reranking achieved 83.2 R@1.

Technical Skills

Programming Languages: C++, Java, Python, TypeScript, JavaScript, SQL
Backend Development: FastAPI, Node.js, Express.js, Spring Boot, REST APIs, Asynchronous Programming, JWT Authentication, BullMQ
Frontend Development: React.js, Next.js, Tailwind CSS, HTML5, CSS3, Jetpack Compose
Databases and Storage: PostgreSQL, MongoDB, Redis, SQLite, Qdrant, Prisma, Firebase, Firestore
Systems and Architecture: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Internals, Query Optimization, Distributed Systems, Concurrency, Caching
AI and Machine Learning: RAG, PyTorch, Hugging Face Transformers, LangChain, Embeddings, NLP, Computer Vision, YOLOv8, ResNet-50
Developer Tools and Testing: Git, GitHub, Linux, Docker, CMake, pytest, Vitest, Playwright, Catch2, Vercel, Render

Achievements

- **Rank 17:** Apple Developer Academy, University of Naples Federico II, Italy.
- **OpenEnv Meta Hackathon:** Placed among the top 800 teams.
- **Finalist:** Smart India Hackathon (SIH 2025).